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~~ASTHMA MORTALITY IN BRAZIL~~

~~Reassessing the reported increase in asthma mortality in Brazil: reproduction and age-specific reanalysis, 2014–2021~~

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~~Reassessing the reported increase in asthma mortality in Brazil: reproduction and age-specific reanalysis, 2014–2021~~

Reassessing asthma mortality in Brazil during 2014–2021 through reproduction and age-specific analysis

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Abstract

~~Objective: To assess whether rising crude asthma mortality in Brazil (2014–2021) persists after age standardisation and to examine trends at ages 5–34.~~

~~Methods: Using published code, national mortality records (ICD-10 J45–J46) and Brazilian Institute of Geography and Statistics (IBGE) projections, we reproduced the original analysis, calculated age-standardised rates (Brazil 2014) and assessed eight-year trends. Count models, an alternative IBGE population projection (2024 revision), World Health Organization (WHO) standardisation and Kitagawa decomposition were post hoc.~~

~~Results: We reproduced 18,583 deaths versus 18,584 reported. The crude national rate rose by 12.5% (slope 0.025 per 100,000/year; p = 0.029). Among completed ages ≥7 (18,544 deaths), the aligned crude rate rose by 12.0%, versus a 4.0% fall in the standardised rate (eight-year slope 0.0002 per 100,000/year; 95% CI −0.021 to 0.022; p = 0.983). Age composition accounted for 62–193% of crude-rate increases across six endpoint comparisons. Only one of six age-adjusted count-model specifications, negative binomial regression with 2024 denominators, yielded a positive estimate whose interval excluded zero (+1.14% annually; 95% CI 0.04 to 2.26). Rates at ages 5–34 increased by 20.3% (p = 0.014), consistently across sensitivity analyses.~~

~~Conclusion: Changes in population age composition accounted for most of the crude increase, without establishing a causal explanation. The primary age-standardised analysis did not support a comparable national increase, whereas mortality at ages 5–34 rose. Surveillance should report crude, age-standardised and age-specific rates jointly.~~

~~Keywords: Asthma; Mortality; Age Distribution; Population Dynamics; Reproducibility of Results; Brazil.~~

Objective. To assess whether rising crude asthma mortality in Brazil during 2014–2021 persists after standardisation at ages ≥7, and examine mortality within ages 5–34.

Methods. This ecological study reproduced the analysis and revised selection using calendar age, COVID exclusions including U071, and retention of two 2018 records without confirmed shared identity. The primary analysis used Brazilian Institute of Geography and Statistics (IBGE) 2018 denominators and Brazil 2014 standard weights,

normalised within each age range. Kitagawa decomposition described the contributions of age composition and age-specific rates to changes in crude mortality. Sensitivities examined IBGE 2024, World Health Organization (WHO) weights, count models, selection rules and pandemic-period restrictions.

Results. Historical selection reproduced 18,583 deaths with continuous age >6; the published count was 18,584. The corrected national analysis of completed ages ≥7 included 18,544 deaths. Aligned crude mortality increased 12.0%; Brazil-2014-standardised mortality changed -4.1% between endpoints (eight-year slope 0.0002; 95% CI -0.0214 to 0.0217; p = 0.986). Within ages 5–34, Brazil-2014-standardised mortality increased 18.0% (0.0328 per 100,000; p = 0.018). Its slope remained positive excluding 2020 (p = 0.040); the 2014–2019 interval included zero (p = 0.107).

Conclusions. Changes in population age composition accounted for much of the crude increase at ages ≥7 in a descriptive decomposition. The primary national age-standardised estimate was compatible with trends in either direction. Within 5–34, primary Brazil-2014 standardisation supported an increase over 2014–2021 and after excluding 2020; the shorter prepandemic series remained inconclusive about the annual trend. Surveillance should report crude, age-standardised and age-specific rates together.

Keywords. Asthma; Mortality; Age Distribution; Population Dynamics; Reproducibility of Results; Brazil.

Introduction

~~Brum et al. concluded that “asthma mortality remains high and shows an increasing trend for the first time in the past decades” [1]. Their nationwide analysis for 2014–2021 identified 18,584 asthma deaths and reported an approximately 2.5% annual increase in asthma mortality. A subsequent study by Cardoso et al., covering 2013–2023, also concluded that asthma-related deaths had increased [2]. However, this conclusion was based on annual death counts, while the same study reported that the overall mortality rate per 100,000 inhabitants remained stable. Thus, the two studies described increasing asthma mortality using different measures, neither of which established an increase in the national age-standardised rate.~~

~~Earlier Brazilian surveillance provides a longer perspective. Graudenz et al. reported a linear decline in age-standardised asthma mortality among Brazilians aged 5–34 years during 1980–2014, whereas the trend in the general population was non-linear [3]. Differences in study periods, age ranges and rate definitions prevent direct comparison with more recent estimates or the identification of a historical trend reversal.~~

~~Neither Brum et al. nor Cardoso et al. directly age-standardised the national mortality series. This omission matters because approximately two thirds of asthma deaths occurred among individuals aged 60 years or older [1,2], and changes in population age structure can affect crude mortality rates independently of changes in age-specific mortality. Consequently, an increase in deaths or crude mortality cannot, by itself, establish a comparable increase in age-adjusted mortality. The public availability of Brum et al.’s analytical code [4] also makes it possible to examine the reproducibility of their case selection, population denominators and trend estimates before reassessing the interpretation.~~

~~We therefore asked to what extent the reported increase in asthma mortality in Brazil during 2014–2021 remains supported when the original analysis is reproduced and population age structure is held fixed. We first reproduced Brum et al.’s analysis and documented agreements and discrepancies with the published results. We then estimated national age-standardised mortality trends and examined mortality among individuals aged 5–34 years. In post hoc robustness analyses, we assessed an alternative population projection, the WHO standard population and count regression models. Our objective was not to dispute the observed increase in crude mortality, but to establish which epidemiological interpretations are supported after accounting for population age structure and examining age-specific trends.~~

One nationwide study concluded that "asthma mortality remains high and shows an increasing trend for the first time in the past decades" [1]. That analysis for 2014–2021 identified 18,584 asthma deaths and reported an annual increase of approximately 2.5%. Another study of 2013–2023 concluded that asthma-related deaths had increased [2]. The latter conclusion relied on annual death counts, although the overall mortality rate per 100,000 inhabitants was reported as stable. These measures describe different aspects of mortality. Neither establishes an increase in the national age-standardised rate.

Earlier Brazilian surveillance offers a longer perspective. An earlier study reported a linear decline in age-standardised asthma mortality among people aged 5–34 during 1980–2014 and a non-linear trend in the general

population [3]. Differences in periods, age ranges and rate definitions limit direct comparison with recent estimates and prevent identification of a historical trend reversal.

Population ageing is relevant to this comparison because approximately two thirds of asthma deaths occurred among people aged ≥60 [1,2]. Changes in age composition can raise crude mortality even when age-specific rates do not increase. Neither recent study directly age-standardised the national series. Public code for the first study [4] provides an opportunity to examine case selection, population denominators and trend estimates before reassessing the mortality pattern.

We examined whether the reported increase in Brazilian asthma mortality during 2014–2021 persisted when population age structure was held fixed. We first reproduced the first study's analysis and documented agreements and discrepancies with the published results. We then estimated national rates among people aged ≥7 years and rates within ages 5–34, standardising separately within each range. We also used Kitagawa decomposition to describe the contributions of changes in age composition and age-specific rates to the crude-rate change. Post hoc sensitivity analyses examined alternative population projections, WHO weights, count models, selection rules and pandemic-period restrictions.

Methods

Study design

We conducted a secondary analysis of publicly available mortality data, reproducing the national asthma mortality analysis by Brum et al. for 2014–2021 [1] and examining two extensions: direct age standardisation of national mortality rates and mortality trends among individuals aged 5–34 years. The project documentation identifies these as the primary questions, with ages 7–34 years as a technical control and ages 35–59 years as an additional comparison. The Brazil 2014 population was selected as the primary standard. Reproduction was assessed against a tolerance of 2% for the total number of deaths and agreement in the direction and approximate magnitude of the original age-group trends. Count regression models, the WHO standard population and an alternative population projection were subsequently introduced as post hoc robustness analyses.

Mortality data and case selection

Mortality records for 2014–2021 were obtained from the Brazilian Mortality Information System (SIM) [5] on 1 October 2026. The original source URLs were no longer fully accessible, so all years were retrieved from the current official archives. Possible retrospective revisions of these data may affect exact reproduction.

Deaths were selected when the underlying cause contained ICD-10 J45 or J46. Records containing the COVID-related codes B342 or U072, or the text “COVID-19”, in the cause-of-death lines were excluded. Age was calculated from birth and death dates as the elapsed number of days divided by 365.25. Records with invalid or missing birth dates were excluded from age-specific analyses.

Duplicate records were identified using the birth and death dates, underlying cause and five available cause-of-death fields. This procedure uses the variables loaded by our pipeline but does not exactly reproduce the original full-record deduplication.

To reproduce Brum et al., we retained deaths with calculated age greater than six years. The 5–34-year analysis instead used all age-valid records, including deaths at ages five and six.

Population denominators

The original population workbook was unavailable. We reconstructed denominators using IBGE national population projections by single year of age and both sexes combined. The 2018 revision was used in the primary analysis [6], and the 2024 revision, incorporating the 2022 Census, in sensitivity analyses [7]. The exact population series used by Brum et al. could not be recovered.

Mortality rates and age standardisation

For the reproduction of the national crude series, deaths above six years of age were divided by the total population, following the construction inferred from the published code [4]. We also calculated a version restricted to the population above six years. Original age groups were reproduced using their apparent numerator and denominator definitions, with unresolved denominator differences documented in the reproducibility audit.

~~Direct age standardisation used single-year age-specific mortality rates and the corresponding population weights from Brazil in 2014. Ages were represented as completed years from 7 to 89, with ages 90 and above grouped together. Consequently, this analysis used completed ages of at least seven years, whereas the reproduction retained the original continuous-age threshold greater than six years. These definitions were kept distinct rather than treated as identical samples.~~

~~Rates for ages 5–34, 7–34 and 35–59 years used deaths and population denominators restricted to the corresponding ages. Post hoc analyses also used the WHO World Standard Population [8].~~

~~Statistical analysis~~

~~The primary trend models followed those of the published analysis: Gaussian generalised linear regression for national annual rates and ordinary least squares for age-group rates. Eight annual observations were available for each series. Slopes were expressed as deaths per 100,000 population per year, with 95% confidence intervals calculated using the t distribution with six degrees of freedom. We also reported changes between 2014 and 2021. For comparison with Brum et al., we calculated the mean of annual relative differences from the 2014 rate, their reported change metric.~~

~~Post hoc analyses used Poisson, quasi-Poisson and negative binomial regression [9], with population offsets. A national age-adjusted count model included calendar year and nine age bands. Overdispersion was assessed to guide interpretation of count-model uncertainty. In a further post hoc analysis, the change in the crude rate for completed ages of 7 years and older was split into an age-composition component and an age-specific-rate component with the symmetric Kitagawa decomposition, using single years of age and the 2018 projection. Because a comparison between two years depends on its endpoints, the decomposition was repeated for six pairs of years. No correction for multiple comparisons was applied; secondary estimates were interpreted descriptively.~~

~~Software, data and code availability~~

~~Analyses were performed in R 4.3.3 [10]. The analysis code, intermediate aggregate tables, generated results and reproduction audit are available at https://github.com/EduIntellect/asthma-mortality-brazil-reanalysis. Mortality and population data were obtained from public Brazilian sources.~~

~~Use of artificial intelligence~~

~~Generative AI tools, including Claude Code (Anthropic) and GitHub Copilot (GitHub), were used to generate and review analysis scripts and to examine analytical results. The authors wrote and finalised the manuscript.~~

~~The figures were rendered programmatically from analytical outputs rather than generated as images by an AI image-generation tool.~~

~~Ethics~~

~~The study used publicly accessible SIM mortality data and aggregated population projections without direct contact with participants. No identifying personal information was included in the reported results.~~

Study design and reference reproduction

This secondary ecological analysis used the Brazilian Mortality Information System (SIM) to examine national underlying-cause asthma mortality in Brazil during 2014–2021. The corrected primary population comprised people aged ≥7 completed years. We analysed ages 5–34 separately and used the nested 7–34 group to assess sensitivity to ages five and six; 35–59 and ≥60 provided further comparisons. The historical analysis using continuous age >6 was reproduced separately. The primary analysis retained IBGE 2018 denominators and Brazil 2014 standard weights. Sensitivity analyses and descriptive decomposition complemented the primary estimates. We applied no multiple-testing correction.

We reproduced the historical analysis against the archived manuscript and versioned analytical code. Provenance details are reported under Software, reproducibility and ethics.

Selection, COVID mentions and record identity

We selected records whose underlying cause field, CAUSABAS, contained International Classification of Diseases, tenth revision (ICD-10), J45 or J46 [5]. To retain the historical search scope, the corrected COVID rule searched

LINHAA, LINHAB, LINHAC, LINHAD and LINHAII for B342, U072, U071 or literal COVID-19. A match in any field excluded the record once. We logged each code frequency and overlapping combination separately. Sensitivities used the original rule without U071 or retained all records with COVID mentions.

The historical key combined death date, elapsed-days age, underlying cause and five cause lines. It collapsed two 2018 records with differences in occupation and administrative fields. The death-declaration identifier NUMERODO, described in the SIM dictionary [6], is absent from the public CSVs. Available fields therefore cannot establish a shared death. We retained both records and assessed their collapse as a sensitivity. Supplementary methods document the rows and fields.

Calendar ages and analytical domains

Completed age was the difference between death and birth years, reduced by one if death preceded that year's birthday. For births on 29 February, the anniversary in a non-leap year was 1 March. Missing or invalid dates and birth after death were excluded and logged separately. We applied no upper-age exclusion and grouped ages ≥90 into the IBGE 90+ interval. Date parsing is detailed in the supplement.

Calendar age defined the ≥7, 5–34, 7–34, 35–59 and ≥60 groups. One sensitivity used days/365.25, with its floor defining single-year bins. The historical reproduction retained the continuous >6 threshold as a separate population.

Denominators and direct standardisation

National single-year populations for both sexes came from the Brazilian Institute of Geography and Statistics (IBGE) 2018 revision [7]; the 2024 revision following the 2022 Census provided a sensitivity [8]. Both revisions provide single-year ages 0–89 and an open 90+ interval; workbook extraction is documented in the supplement. Corrected crude rates matched each numerator to the same population age range. To reproduce the historical construction, we divided continuous-age >6 deaths by the entire population and separately by population ≥7. The latter comparator retains the small mismatch between its historical numerator and denominator.

For age domain D and year t, we calculated the directly age-standardised rate (ASR) as follows.

$$ASR(t,D) = \Sigma[a \in D]\ w(a,D) \times 100{,}000 \times deaths(a,t) / population(a,t).$$

$$w(a,D) = population(a,2014) / \Sigma[j \in D]\ population(j,2014), \text{ so } \Sigma[a \in D]\ w(a,D) = 1.$$

Weights were fixed across years and normalised separately within each age domain, including 5–34 and 7–34. Each denominator revision supplied its own Brazil 2014 standard population. Deaths and population at ≥90 shared one open interval. World Health Organization (WHO) standardisation used five-year bands [9], combining the 90–94, 95–99 and ≥100 weights for 90+. We allocated three fifths of the WHO 5–9 weight to 7–9 before normalisation, assuming a uniform distribution within that band. Missing death-count cells were assigned zero; population cells had to be complete and positive. We checked that Brazil-2014-standardised and aligned crude rates were equal in 2014.

Trends, count models and sensitivity scenarios

We fitted Gaussian-family generalised linear models (GLMs) to national crude and Brazil-standardised annual rates and ordinary least squares (OLS) to age-group and WHO-standardised rates. The identity-link Gaussian trends are numerically equivalent to OLS. Slopes were expressed as deaths per 100,000 per year, with t intervals using $n-2$ residual degrees of freedom. We reported absolute and relative endpoint changes separately from annual slopes.

Count models used log-link Poisson, quasi-Poisson and negative binomial families [10]. For unadjusted age domains, $\log E[Y(t)] = \alpha + \beta \times year(t) + \log population(t)$. The national age-adjusted model was $\log E[Y(b,t)] = \alpha + \gamma(b) + \beta \times year(t) + \log population(b,t)$. Its nine categorical bands were 7–14 (reference), 15–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75–84 and 85+. Annual percentage change was $APC = 100 \times (\exp \beta - 1)$, with intervals obtained by transforming $\beta \pm$ critical value × standard error (SE). Quasi-Poisson used the t distribution with model residual degrees of freedom; Poisson and negative binomial used normal Wald intervals.

Quasi-Poisson scales Poisson coefficient variance by estimated dispersion. Negative binomial estimates the quadratic variance function $\mu + \mu^2/\theta$ and can change coefficients as well as uncertainty. For every specification, we recorded Pearson dispersion, θ, mean-model convergence, iterations, errors and warnings. Large θ estimates with iteration warnings were interpreted as approaching the Poisson limit; these warnings prevent a claim of successful θ convergence or exact equivalence to Poisson.

Each sensitivity changed one condition from the corrected primary analysis. We examined elapsed-days age, the original COVID rule, retention of all COVID mentions, the historical deduplication key, exclusion of 2020 and restriction to 2014–2019. The last two focused on 5–34 and 7–34 and retained actual calendar years as covariates. Excluding 2020 left seven observations and 5 trend residual degrees of freedom; restriction to 2014–2019 left six and 4. We evaluated both denominator revisions and both standards and reported every model regardless of statistical significance.

Kitagawa accounting decomposition

For the corrected ≥7 population, symmetric Kitagawa decomposition [11] partitioned crude differences into an age-composition term C and an age-specific-rate term R.

$C = \Sigma[a]\ (w(a,1) - w(a,0)) \times (r(a,0) + r(a,1)) / 2.$

$R = \Sigma[a]\ (r(a,1) - r(a,0)) \times (w(a,0) + w(a,1)) / 2.$

Crude(1) − crude(0) = C + R, with r in deaths per 100,000 and each year's w normalised within ≥7.

We used the same six endpoint pairs with IBGE 2018 and 2024, namely 2014–2021, 2014–2019, 2014–2020, 2015–2021, 2016–2021 and 2014–2018. We checked the identities at full precision before rounding each displayed component independently. This can produce a discrepancy in the last displayed digit. Shares above 100% indicate that a negative rate term offsets part of the composition term. The decomposition is descriptive and does not identify causal mechanisms.

Software, reproducibility and ethics

We used R 4.6.1 [12] for reproduction and revision; the reference reported R 4.3.3. The code records versions, configurations, hashes, warnings and independent checks. Public inputs and separate scenario outputs support reproduction. Code and revision outputs are available at https://github.com/EduIntellect/asthma-mortality-brazil-reanalysis/tree/revision/amaury-review .

Recovered official SIM archives reproduced the recorded selection counts and year-age aggregates exactly. Historical execution hashes exist only for 2020–2021, preventing confirmation of historical file identity for 2014–2019. The supplement and provenance manifest document the manuscript reference, concordant Word, code commit and input hashes.

Codex assisted code revision and analysis checking. Figures were plotted directly from analytical results. Editorial AI assistance is disclosed in Acknowledgements.

The study used public SIM mortality records and aggregate IBGE projections, with no participant contact, and reports only aggregate results. No formally granted ethics approval or exemption is claimed.

Results

~~Reproduction of the original analysis~~

~~The SIM files for 2014–2021 contained 19,105 deaths with asthma as the underlying cause. After the COVID-19-related exclusion and removal of duplicates (57 records, 56 of them in 2020–2021) and exclusion of 15 deaths without a valid date of birth, the reproduction included 18,583 deaths above 6 years of age, against 18,584 reported by Brum et al. [1] (Supplementary Table S1). Of these, 68.2% occurred at 60 years or older, as in the original study.~~

~~The crude national rate rose from 1.01 to 1.14 per 100,000 between 2014 and 2021 (+12.5%), with a maximum of 1.26 in 2020 (Supplementary Table S2). The linear trend was 0.025 per 100,000 per year (95% CI 0.004 to 0.046; p = 0.029; Table 1), against 0.03 (95% CI 0.01 to 0.04; p = 0.01) in the original paper. A negative binomial model gave an annual increase of 2.3% (95% CI 1.0 to 3.6), close to the 2.5% reported originally. The published annual rates were 1.8% to 5.4% higher than ours, with a gap that widened over the period, which points to a different population series; we could not identify it.~~

~~For the age groups of the original analysis, our slopes agreed in sign and significance class with the published ones: no trend below 18 years (p = 0.85), an increase at 18–59 years (slope 0.021; p = 0.044) and a non-significant decrease at 60 years or older (slope −0.071; p = 0.14). The magnitude of the changes did not match exactly, and the largest difference, at 60 years or older, could not be reconciled (Supplementary Table S3). The reproduction thus met~~

~~the criteria defined in the project documentation: total deaths within 2% and agreement in the sign of the age-group trends.~~

~~National trend after age standardisation~~

~~Using the same completed-age population of 7 years and older for both calculations, the crude rate increased from 1.123 to 1.258 per 100,000 between 2014 and 2021 (+12.0%), whereas the directly age-standardised rate decreased from 1.123 to 1.078 (−4.0%; Table 1; Figure 1). The estimated slope of the age-standardised series over all eight years was 0.0002 per 100,000 per year (95% CI −0.0213 to 0.0217; p = 0.983). Thus, the analysis did not support an increasing national age-standardised trend, although the confidence interval did not establish equivalence to a zero trend.~~

~~In post hoc count models (Supplementary Table S4), the age-adjusted national trend was +0.00% per year with quasi-Poisson intervals (95% CI −1.12 to 1.14) and +0.9% (−0.2 to 2.1; p = 0.10) with a negative binomial model. With the 2024 population projection, the negative binomial estimate was +1.1% per year (95% CI 0.04 to 2.3; p = 0.043), the only one of six age-adjusted specifications whose interval excluded zero. The crude series was overdispersed (dispersion 5.9); its annual increase of 2.3% had p = 0.029 with quasi-Poisson and p < 0.001 with negative binomial intervals. Using the WHO standard population, the standardised rate changed by −4.2% (p = 0.99) with the 2018 projection and by −2.0% (p = 0.73) with the 2024 projection (Supplementary Table S5).~~

~~Decomposition of the change in the crude rate~~

~~The age-composition component accounted for 62% to 193% of the observed crude-rate increase across the six endpoint comparisons, while the component associated with age-specific mortality rates changed sign (Figure 2; exact values in Supplementary Table S6). For 2014–2021, the positive composition effect (+0.191 per 100,000) exceeded the net increase (+0.135) and was partly offset by changes in age-specific rates (−0.057). Approximately 95% of the composition component arose among individuals aged 60 years or older.~~

~~Ages 5–34 years~~

~~The rate at ages 5–34 years rose from 0.182 to 0.219 per 100,000 between 2014 and 2021 (+20.3%; slope 0.0080 per year, 95% CI 0.0023 to 0.0138; p = 0.014; Table 1). The group had 1,591 deaths over the period, between 172 and 236 per year, with the lowest rate in 2017 (0.175) and the highest in 2020 (0.244) (Supplementary Figure S1). The control group aged 7–34 years gave a similar result (+23.5%; p = 0.011; Supplementary Figure S1); it contains nearly all the deaths of the 5–34 group, so it shows that the estimate does not depend on ages 5 and 6 rather than replicating it independently.~~

~~The increase persisted in post hoc analyses. Overdispersion was negligible (1.07), and the quasi-Poisson model gave an annual increase of 4.0% (95% CI 1.2 to 7.0; p = 0.013); with the 2024 projection it was 4.2% (p = 0.011). Standardised to the WHO population within the group, the rate rose by 16.9% (p = 0.020) with the 2018 projection and by 19.1% (p = 0.015) with the 2024 projection.~~

~~At 35–59 years the evidence was inconclusive: the rate rose from 0.719 to 0.830 (+15.5%; p = 0.13), and the count models disagreed (p = 0.13 with quasi-Poisson and p = 0.036 with negative binomial).~~

Historical reproduction and corrected selection

The historical pipeline reproduced 19,105 underlying-cause candidates, 56 COVID exclusions, one reduced-key removal, 15 uninterpretable ages and 18,583 deaths at continuous age >6. The original study reported 18,584 deaths [1]. All versioned selection counts, annual age aggregates and IBGE-2018 denominators matched exactly. Rate and trend differences were at floating-point scale. Negative binomial θ estimates differed slightly under the current environment in models approaching the Poisson limit, with no change in significance classifications. Supplementary Table S3 documents the unresolved original population series and age-group discrepancies.

The eight official files contained 11,169,644 records, including 19,105 underlying-cause asthma records. Corrected COVID exclusions removed 57 records, comprising 27 in 2020 and 30 in 2021. No record was removed as a proven duplicate; 15 had non-interpretable dates and none had birth after death, leaving 19,033 age-valid records. The ≥7 domain contained 18,544 deaths, including 12,678 (68.4%) at ≥60. Calendar age changed the single-year bin of 23 candidates (23 retained), but none crossed the ≥7, 5–34, 7–34, 35–59 or ≥60 domain boundaries relative to

floor(days/365.25). Ages above 90 were retained in the 90+ interval. The corrected total ≥7 equals the historical total because retaining one 2018 record and excluding the additional 2021 U071 record cancel in the total.

National domain of completed ages ≥7

For completed ages ≥7, the aligned crude series rose from 1.123 to 1.257 per 100,000 (+12.0%; absolute change +0.1343), with slope 0.0270 per year (95% CI 0.0030 to 0.0509; p = 0.033). Brazil-2014-standardised mortality changed from 1.123 to 1.077 per 100,000 (-4.1%; absolute change -0.0456), with slope 0.0002 per year (95% CI -0.0214 to 0.0217; p = 0.986). The endpoint comparison describes the difference between 2014 and 2021. The slope uses all eight annual observations, and its interval includes trends in both directions.

Table 1 summarises the rate estimates. Figure 1 compares changes from 2014.

With 2018 denominators, the age-adjusted national estimate was +0.00% per year (95% CI -1.13 to 1.14; p = 0.999) under quasi-Poisson and +0.93% per year (95% CI -0.20 to 2.06; p = 0.106) under negative binomial regression. With 2024 denominators, the negative binomial estimate was +1.14% per year (95% CI 0.03 to 2.26; p = 0.044), the only one of the six corrected national age-adjusted specifications whose interval excluded zero. The national Poisson Pearson dispersion was 3.11; different uncertainty and variance assumptions contribute to differences between families. All mean-model iterations converged. Some unadjusted age-group negative binomial fits reached the theta iteration limit. Supplementary Table S4 identifies these warnings; convergence of the mean model alone does not establish dispersion convergence.

Composition of the crude-rate change at ≥7

Across six pairs, the composition component was positive under both revisions. With IBGE 2018 it accounted for 61.5–193.8% of the net increase, compared with 55.0–153.5% under IBGE 2024. For 2014–2021, the 2018 net change of +0.134 per 100,000 comprised composition +0.191 and age-specific rates -0.057; 94.8% of composition arose at ≥60. The corresponding 2024 terms were +0.146, +0.170 and -0.025. Contributions above 100% mean that a negative rate component partly offset the positive composition component; these proportions describe the decomposition and do not estimate causal effects.

Figure 2 compares the decomposition under both population revisions.

Age-domain mortality and sensitivity analyses

Ages 5–34 contributed 1,590 corrected deaths (172–236 annually). Their crude rate rose from 0.182 to 0.218 per 100,000 (+19.7%; absolute change +0.0358), with slope 0.0079 per year (95% CI 0.0021 to 0.0138; p = 0.016). Standardising within 5–34 to Brazil 2014 yielded 0.182 to 0.215 per 100,000 (+18.0%; absolute change +0.0328), with slope 0.0075 per year (95% CI 0.0018 to 0.0132; p = 0.018); WHO standardisation yielded 0.178 to 0.207 per 100,000 (+16.4%; absolute change +0.0291), with slope 0.0068 per year (95% CI 0.0014 to 0.0122; p = 0.022). Thus the crude absolute increase was 0.0358, and the within-group-standardised absolute increase was 0.0328 per 100,000. The nested 7–34 comparison gave 0.181 to 0.220 per 100,000 (+21.0%; absolute change +0.0381), with slope 0.0088 per year (95% CI 0.0026 to 0.0149; p = 0.013); it tests sensitivity to including ages five and six. The crude 5–34 quasi-Poisson estimate was +4.00% per year (95% CI 1.13 to 6.95; p = 0.014).

Excluding 2020 while retaining calendar-year values, the within-5–34 Brazil-2014-standardised slope was 0.0058 (95% CI 0.0004 to 0.0112; p = 0.040). In 2014–2019 it was 0.0068 (95% CI -0.0023 to 0.0158; p = 0.107); the endpoint increase was 22.4%. For the 7–34 control, the respective standardised trend p values were 0.021 and 0.067. The primary standardised prepandemic interval therefore does not establish a positive annual trend.

With WHO standardisation and IBGE 2018, excluding 2020 gives p = 0.055, with an interval including zero. In 2014–2019, count-model inference also differs. The crude 5–34 Poisson and negative binomial Wald intervals exclude zero; the quasi-Poisson t interval includes it; negative-binomial theta has an iteration warning. Supplementary Table S7 reports results for both revisions, both standards and all three count families. These comparisons show that inference depends on the standard and variance assumptions.

The original COVID rule removes 56 records; adding U071 removes one further record in 2021, within both young domains. Of 18 U071 mentions in 2020, all were already excluded; of 19 in 2021, 18 overlapped the original rule. Retaining all COVID mentions raises the primary ≥7 numerator from 18,544 to 18,601. Calendar-age, original-

COVID, all-COVID-retained and historical-key sensitivities change one condition at a time (Supplementary Table S8).

At 35–59 the crude slope remained imprecise (p = 0.132); uncertainty is not evidence of stability.

Discussion

We reproduced the crude increase in asthma mortality reported by Brum et al. for 2014–2021, with one death of difference in the total count and the same direction of the national and age-group trends. After direct age-standardisation, however, the national age-standardised trend was not statistically distinguishable from zero: the standardised rate fell by 4.0% and its slope over the eight years was close to zero, with a confidence interval that admits small changes in either direction. In a decomposition of the crude change, shifts in the age composition of the population accounted for most of the increase, and in the 2014–2021 comparison for more than all of it. The rise in mortality at 5–34 years, by contrast, persisted across every specification we examined.

These results do not contradict the crude series of Brum et al.; they qualify its interpretation. A rising crude rate measures a rising burden of deaths in the population, which is relevant for health services. It does not by itself show that the risk of dying from asthma at a given age has increased. Cardoso et al. reported an increase in the number of asthma deaths over 2013–2023 while noting that the overall rate per 100,000 inhabitants remained stable [2]. Together, these studies illustrate why trends in death counts and crude rates should be distinguished from age-adjusted trends. In our reanalysis, changes in population age composition accounted for much of the crude increase. The statement that asthma mortality "shows an increasing trend for the first time in the past decades" [1] is consistent with the crude-rate increase reported by Brum et al., but our analysis does not establish a comparable increase in national age-standardised mortality during 2014–2021.

The Kitagawa decomposition gives this reading a quantitative form, but it is an accounting decomposition, not a causal one. It splits a difference between two years into a component due to the age distribution and a component due to age-specific rates; it does not identify why either changed. Because two-year comparisons depend on their endpoints, we repeated it for six pairs of years. The composition component was positive in all of them, and about 95% of it arose at 60 years or older, the age range whose share of the population grew from 13.3% to 16.3%. The age-specific component changed sign with the endpoints, and was largest in the comparison ending in 2020, the year with the highest rates. We therefore treat the eight-year standardised trend as the main evidence, with the decomposition describing contributions to the crude increase.

The national age-adjusted result was not uniform across specifications. One of six count-model specifications, the negative binomial model with the 2024 population projection, estimated an increase of 1.1% per year whose interval excluded zero. The remaining specifications gave estimates between 0% and 0.9% per year with intervals that included zero. With eight annual observations, these data cannot establish that age-adjusted mortality was constant; the age-adjusted point estimates were generally smaller than the crude estimate, although the confidence intervals did not rule out increases of comparable magnitude under some specifications.

The increase at ages 5–34 requires separate interpretation. The rate rose by about 20% between 2014 and 2021, and the estimate held with count models, with the WHO standard population and with the alternative projection. This age range was used in earlier Brazilian surveillance, in which Graudenz et al. reported a decline in age-standardised mortality from 1980 to 2014 [3]. Our primary estimate for this group is not standardised within the group, and the two studies differ in period and methods, so the comparison does not establish a change point in 2014. The finding rests on 172 to 236 deaths per year, and the highest rate occurred in 2020. It warrants monitoring with longer series rather than a firm conclusion about a reversal.

The reproduction also revealed limits to exact comparison with published estimates. Our population denominators differed from those of the original study, so our annual national crude rates were 1.8% to 5.4% lower, and the magnitude of the change at 60 years or older could not be reconciled. The original code made it possible to identify these differences, but the exact population series could not be recovered. Reporting the denominators used, together with age-standardised rates alongside crude rates, would make national asthma surveillance comparable across studies and years.

Several limitations remain. The design is ecological and descriptive. The count models, the WHO standard population, the alternative projection and the decomposition were added after the main analyses and should be read

~~as robustness or descriptive analyses, and we made no adjustment for multiple comparisons. The SIM files may have been revised since the original analysis, our deduplication did not reproduce the original procedure exactly, and changes in death certification or coding during the period, including during the COVID-19 pandemic, cannot be excluded. The questions were defined in the project documentation, but we could not document their timing relative to the analyses independently.~~

~~Overall, the increase in crude asthma mortality in Brazil between 2014 and 2021 is reproducible, but it was not accompanied by a comparable increase in age-standardised mortality, and changes in population age structure accounted for most of it. Mortality at 5–34 years did increase over the period. National surveillance should report age-standardised and age-specific trends together with crude rates, because in this case they lead to different conclusions about the direction of asthma mortality.~~

The corrected analysis found an increase in aligned crude asthma mortality among people aged ≥7 between 2014 and 2021. With Brazil 2014 age weights held fixed, the endpoint change was negative and the slope across all eight years was close to zero. Endpoint comparisons and slopes across all annual observations describe different aspects of the series. The interval was compatible with trends in either direction and does not demonstrate stability. The positive age-adjusted negative binomial estimate with IBGE 2024 shows that national trend inference also depends on the model and population revision.

Age composition contributed positively to the crude change for every endpoint pair under both population revisions. In some pairs, the composition term exceeded the net increase because the age-specific-rate term was negative. The decomposition combines composition and rates arithmetically; it cannot identify effects of certification, treatment access or the pandemic.

Both crude and standardised mortality within ages 5–34 increased over 2014–2021. Standardising within this range accounts for changes in its internal age distribution. The nested 7–34 analysis assesses how including ages five and six affects the estimate and does not provide independent replication.

Extension of the finding across periods depends on the analysis. The primary Brazil-standardised slope retained a positive interval after excluding 2020; the corresponding WHO sensitivity with IBGE 2018 included zero. The shorter prepandemic primary estimate was imprecise and its interval included zero. Its positive endpoint contrast alone does not establish an annual trend over 2014–2019. The observed increase over 2014–2021 warrants continued surveillance, while differences in study periods and age definitions prevent identification of a change point or reversal of the earlier decline reported previously [3].

Interpretation also depends on which records enter each analysis. The selection audit shows why equal ≥7 totals need not represent the same records. Retaining one ambiguous 2018 record and excluding one additional U071 record in 2021 cancel in the total. Calendar age also changed single-year bins without changing membership in the five age domains, affecting the age distribution used in standardisation and count models. Separate sensitivities identify the effect of each alternative selection rule.

The ecological design, short annual series, post hoc sensitivity analyses, unadjusted multiple comparisons and possible changes in death certification limit interpretation. Public fields cannot establish whether the 2018 pair represents one death. Historical execution hashes for SIM 2014–2019 and the population workbook used in the original study remain unavailable. Exact reproduction of recorded counts and aggregates therefore cannot establish historical file identity.

Negative binomial θ iteration warnings near the Poisson limit constrain confidence in dispersion estimation. We retained every model and its warnings to make these limitations visible. Statistical intervals quantify model uncertainty; they do not capture uncertainty in record selection, certification or population reconstruction.

These findings support reporting aligned crude, age-standardised and age-specific mortality together. Age composition accounted for much of the crude increase among people aged ≥7. The increase within 5–34 over 2014–2021 merits attention, with interpretation bounded by prepandemic uncertainty and sensitivity to model assumptions.

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Conflicts of interest

The authors declare no conflicts of interest.

Author contributions

Francisco Federico García Crespi contributed conceptualisation, methodology, software, formal analysis, data curation, visualisation and writing the original draft. Amaury de Souza contributed validation of epidemiological and scientific interpretation, and writing, review and editing. Both authors approved the final manuscript.

References

Brum M, Henz J, Boeira M, Soares S, Friedrich F, Pitrez PM. Recent increase in asthma mortality in Brazil: a warning sign for the public health system. J Bras Pneumol. 2024;50(5):e20240138. doi:10.36416/1806-3756/e20240138

Cardoso ADRO, Ferreira ACG, Rabahi MF. Asthma-related deaths in Brazil: data from an ecological study. J Bras Pneumol. 2024;50(6):e20240296. doi:10.36416/1806-3756/e20240296

Graudenz GS, Carneiro DP, Vieira RP. Trends in asthma mortality in the 0- to 4-year and 5- to 34-year age groups in Brazil. J Bras Pneumol. 2017;43(1):24-31. doi:10.1590/S1806-37562015000000253

Brum M, et al. Analysis code accompanying reference 1. GitHub repository, commit 8d527a5 (15 May 2024). Available from: https://github.com/mobrant94/asthma_mortality

Ministério da Saúde (Brasil). Sistema de Informação sobre Mortalidade (SIM). Portal de Dados Abertos do SUS. Available from: https://dadosabertos.saude.gov.br/dataset/sim

Instituto Brasileiro de Geografia e Estatística. Projeção da População do Brasil e das Unidades da Federação, revisão 2018: população por idade simples, 2010–2060. Available from: https://ftp.ibge.gov.br/Projecao_da_Populacao/Projecao_da_Populacao_2018/ (accessed 1 October 2026)

Instituto Brasileiro de Geografia e Estatística. Projeções da População, revisão 2024: população por idade simples. Available from: https://ftp.ibge.gov.br/Projecao_da_Populacao/Projecao_da_Populacao_2024/ (accessed 1 October 2026)

Ahmad OB, Boschi-Pinto C, Lopez AD, Murray CJL, Lozano R, Inoue M. Age standardization of rates: a new WHO standard. GPE Discussion Paper Series No. 31. Geneva: World Health Organization; 2001.

Venables WN, Ripley BD. Modern applied statistics with S. 4th ed. New York: Springer; 2002.

R Core Team. R: a language and environment for statistical computing. Vienna: R Foundation for Statistical Computing; 2024. Available from: https://www.R-project.org/

Tables

Table 1. Trend estimates, Brazil, 2014–2021. aChange in the rate, per 100,000 per year, from a Gaussian-family generalised linear model (crude and age-standardised national series) or ordinary least squares (age groups), eight annual observations, 95% confidence intervals from the t distribution with 6 degrees of freedom. Change (%) is the relative change between 2014 and 2021. The two crude rows reproduce the original analysis (deaths above 6 years of continuous age) and differ only in the population base; the age-standardised row uses completed ages of 7 years or more. Population: IBGE 2018 projection.

Series | Rate 2014 | Rate 2021 | Change (%) | Slopea | 95% CI | p

Crude national rate | 1.012 | 1.138 | +12.5 | 0.0250 | (0.0036 to 0.0463) | 0.029

Crude national rate, population >6 years | 1.126 | 1.260 | +11.9 | 0.0268 | (0.0031 to 0.0505) | 0.032

Age-standardised (Brazil 2014) | 1.123 | 1.078 | −4.0 | 0.0002 | (−0.0213 to 0.0217) | 0.983

Ages 5–34 | 0.182 | 0.219 | +20.3 | 0.0080 | (0.0023 to 0.0138) | 0.014

~~Ages 7–34 | 0.181 | 0.224 | +23.5 | 0.0094 | (0.0031 to 0.0156) | 0.011~~

~~Ages 35–59 | 0.719 | 0.830 | +15.5 | 0.0221 | (−0.0089 to 0.0532) | 0.132~~

~~Figures~~

~~Figure 1. Relative changes in asthma mortality rates~~

~~Relative change in each rate from its 2014 value, Brazil, 2014–2021: crude rate (dashed line, open circles) and age-standardised rate (Brazil 2014 standard; solid line, filled circles), both for completed ages of 7 years and older and the same deaths and population, and rate at ages 5–34 years (dash-dot line, triangles). Annual data; population from the IBGE 2018 projection.~~

~~Figure 2. Kitagawa decomposition of crude-rate change~~

~~Kitagawa decomposition of the change in the crude rate, completed ages 7 years and older, for six pairs of years (post hoc analysis), Brazil. Black bars: age-composition component; hatched bars: age-specific-rate component. The two components sum to the change in the crude rate. Deaths per 100,000; population from the IBGE 2018 projection. The components are accounting terms, not causal effects. Exact values in Supplementary Table S6.~~

1. Brum M, Henz J, Boeira M, Soares S, Friedrich F, Pitrez PM. Recent increase in asthma mortality in Brazil: a warning sign for the public health system. J Bras Pneumol. 2024;50(5):e20240138. doi:10.36416/1806-3756/e20240138

2. Cardoso ADRO, Ferreira ACG, Rabahi MF. Asthma-related deaths in Brazil: data from an ecological study. J Bras Pneumol. 2024;50(6):e20240296. doi:10.36416/1806-3756/e20240296

3. Graudenz GS, Carneiro DP, Vieira RP. Trends in asthma mortality in the 0- to 4-year and 5- to 34-year age groups in Brazil. J Bras Pneumol. 2017;43(1):24-31. doi:10.1590/S1806-37562015000000253

4. Brum M, et al. Analysis code accompanying reference 1. GitHub repository, commit 8d527a5 (15 May 2024). Available from: https://github.com/mobrant94/asthma_mortality

5. Ministério da Saúde (Brasil). Sistema de Informação sobre Mortalidade (SIM). Portal de Dados Abertos do SUS. Available from: https://dadosabertos.saude.gov.br/dataset/sim

6. Ministério da Saúde. Estrutura de Dados SIM. Secretaria de Vigilância em Saúde. Official data dictionary, archived by UFBA. https://ccvisat.ufba.br/wp-content/uploads/2019/10/Estrutura-do-SIM-para-o-CD-ROM.pdf (accessed 11 October 2026).

7. Instituto Brasileiro de Geografia e Estatística. Projeção da População do Brasil e das Unidades da Federação, revisão 2018: população por idade simples, 2010–2060. Available from: https://ftp.ibge.gov.br/Projecao_da_Populacao/Projecao_da_Populacao_2018/ (accessed 1 October 2026)

8. Instituto Brasileiro de Geografia e Estatística. Projeções da População, revisão 2024: população por idade simples. Available from: https://ftp.ibge.gov.br/Projecao_da_Populacao/Projecao_da_Populacao_2024/ (accessed 1 October 2026)

9. Ahmad OB, Boschi-Pinto C, Lopez AD, Murray CJL, Lozano R, Inoue M. Age standardization of rates: a new WHO standard. GPE Discussion Paper Series No. 31. Geneva: World Health Organization; 2001.

10. Venables WN, Ripley BD. Modern applied statistics with S. 4th ed. New York: Springer; 2002.

11. Kitagawa EM. Components of a difference between two rates. J Am Stat Assoc. 1955;50(272):1168–1194. doi:10.1080/01621459.1955.10501299

12. R Core Team. R: a language and environment for statistical computing. Vienna: R Foundation for Statistical Computing; 2026. Available from: https://www.R-project.org/

Table 1. Historical comparators and corrected primary rates, Brazil, 2014–2021

Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p

Corrected crude ≥7 | 1.123 | 1.257 | +0.134 | +12.0 | 0.0270 (0.0030, 0.0509) | 0.033

Brazil 2014 ASR ≥7 | 1.123 | 1.077 | -0.046 | -4.1 | 0.0002 (-0.0214, 0.0217) | 0.986

Crude 5–34 | 0.182 | 0.218 | +0.036 | +19.7 | 0.0079 (0.0021, 0.0138) | 0.016

Brazil 2014 ASR 5–34 | 0.182 | 0.215 | +0.033 | +18.0 | 0.0075 (0.0018, 0.0132) | 0.018

Crude 7–34 | 0.181 | 0.223 | +0.042 | +22.9 | 0.0093 (0.0029, 0.0156) | 0.012

Crude 35–59 | 0.719 | 0.830 | +0.111 | +15.5 | 0.0221 (-0.0089, 0.0532) | 0.132

Historical >6 / all population | 1.012 | 1.138 | +0.126 | +12.5 | 0.0250 (0.0036, 0.0463) | 0.029

Historical >6 / ≥7 population | 1.126 | 1.260 | +0.134 | +11.9 | 0.0268 (0.0031, 0.0505) | 0.032

Rates and absolute changes are per 100,000. Slopes are per 100,000 per year with t-based 95% intervals, six residual degrees of freedom. Δ is the endpoint difference; Δ % is the relative endpoint change. Corrected rows use calendar age, U071-inclusive COVID exclusions and retention of both ambiguous 2018 records. Historical rows retain continuous >6 and their original denominators. IBGE 2018 is used throughout this table; primary ASR uses Brazil 2014 weights normalised inside the specified domain.

Figure 1. Relative changes from 2014

Figure 1. Corrected aligned crude and Brazil-2014-standardised national rates for completed ages ≥7, plus rates standardised within 5–34. IBGE 2018 denominators; changes relative to each series' own 2014 rate.

Figure 2. Kitagawa sensitivity to population revision

Figure 2. Corrected ≥7 crude-rate changes partitioned into age-composition and age-specific-rate components for six pairs, with IBGE 2018 and 2024. Components sum to net change before rounding and are not causal effects.

Supplementary material

~~Reassessing the reported increase in asthma mortality in Brazil: reproduction and age-specific reanalysis, 2014–2021~~

~~Supplementary Table S1. Death records by year and stage of case selection. aAfter the COVID-19-related exclusion (B34.2, U07.2 or "COVID-19" in the cause-of-death lines) and removal of duplicate records. The last column is the sample of the reproduction: asthma deaths above 6 years of age with a valid date of birth.~~

~~Year | All records | Asthma (J45–J46) | After exclusionsa | Valid birth date | Age >6 years~~

~~2014 | 1,227,039 | 2,129 | 2,129 | 2,127 | 2,041~~

~~2015 | 1,264,175 | 2,232 | 2,232 | 2,231 | 2,148~~

~~2016 | 1,309,774 | 2,249 | 2,249 | 2,247 | 2,190~~

~~2017 | 1,312,663 | 2,477 | 2,477 | 2,476 | 2,423~~

~~2018 | 1,316,719 | 2,307 | 2,306 | 2,302 | 2,249~~

~~2019 | 1,349,801 | 2,488 | 2,488 | 2,486 | 2,426~~

~~2020 | 1,556,824 | 2,735 | 2,708 | 2,705 | 2,678~~

~~2021 | 1,832,649 | 2,488 | 2,459 | 2,459 | 2,428~~

~~Total | 11,169,644 | 19,105 | 19,048 | 19,033 | 18,583~~

~~Supplementary Table S2. Deaths (upper panel) and rates per 100,000 (lower panel) by year. The crude national rate uses deaths above 6 years of age and the total population; the age-standardised rate uses the 2014 Brazilian age distribution within completed ages of 7 or more. Rates for the age groups are per 100,000 persons of that age. Population: IBGE 2018 projection.~~

~~Series | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021~~

~~Deaths | | | | | | | |~~

~~Crude national rate | 2,041 | 2,148 | 2,190 | 2,423 | 2,249 | 2,426 | 2,678 | 2,428~~

Age-standardised (Brazil 2014) | 2,035 | 2,142 | 2,187 | 2,418 | 2,243 | 2,419 | 2,677 | 2,423

Ages 5–34 | 182 | 189 | 184 | 172 | 200 | 218 | 236 | 210

Ages 7–34 | 171 | 174 | 179 | 164 | 190 | 208 | 232 | 202

Ages 35–59 | 452 | 485 | 515 | 567 | 504 | 524 | 709 | 591

Rate per 100,000 | | | | | | | |

Crude national rate | 1.012 | 1.056 | 1.067 | 1.172 | 1.079 | 1.154 | 1.265 | 1.138

Age-standardised (Brazil 2014) | 1.123 | 1.144 | 1.132 | 1.209 | 1.089 | 1.140 | 1.230 | 1.078

Ages 5–34 | 0.182 | 0.190 | 0.186 | 0.175 | 0.204 | 0.224 | 0.244 | 0.219

Ages 7–34 | 0.181 | 0.186 | 0.192 | 0.177 | 0.207 | 0.228 | 0.256 | 0.224

Ages 35–59 | 0.719 | 0.756 | 0.787 | 0.851 | 0.743 | 0.759 | 1.011 | 0.830

Supplementary Table S3. Age groups as constructed in the original analysis, reproduced with the 2018 projection. aLabelled "below 18" in the original analysis; the numerator contains deaths at 7–17 years and the denominator the population aged 0–17. bMean over 2015–2021 of the relative difference from the 2014 rate, which is the quantity reported by the original analysis. Rates per 100,000 in the stated age group.

Group | Deaths 2014 / 2021 | Rate 2014 / 2021 | Change 2014–2021 (%) | Mean relative variationb (%) | Slope | p

Ages 7–17a | 43 / 37 | 0.076 / 0.069 | −9.2 | +12.1 | −0.0005 | 0.852

Ages 18–59 | 580 / 756 | 0.478 / 0.587 | +22.9 | +16.5 | 0.0212 | 0.044

Ages ≥60 | 1,412 / 1,630 | 5.853 / 5.203 | −11.1 | −1.9 | −0.0712 | 0.135

Supplementary Table S4. Count models with population offset (post hoc analysis), 2018 projection unless stated. aPearson chi-square divided by degrees of freedom in the Poisson model; values well above 1 indicate overdispersion. The age-adjusted model uses nine age bands as a factor. For ages 5–34 and 7–34 the negative binomial model is equivalent to the Poisson model. bAnnual change (%) from quasi-Poisson models with the 2024 projection, with the p value in parentheses.

Series | Dispersiona | Quasi-Poisson, annual change (%) | p | p, neg. binomial | 2024 denominatorsb

Crude national rate | 5.91 | +2.25 (0.32 to 4.22) | 0.029 | <0.001 | +2.40 (0.022)

Age-adjusted, ages ≥7 | 3.11 | +0.00 (−1.12 to 1.14) | 0.994 | 0.104 | +0.29 (0.602)

Ages 5–34 | 1.07 | +4.04 (1.20 to 6.96) | 0.013 | <0.001 | +4.22 (0.011)

Ages 7–34 | 1.13 | +4.65 (1.64 to 7.75) | 0.009 | <0.001 | +4.83 (0.007)

Ages 35–59 | 5.54 | +2.78 (−1.08 to 6.79) | 0.130 | 0.036 | +2.84 (0.123)

Supplementary Table S5. Sensitivity to the standard population and to the population projection (post hoc analysis). For each row the slope is in rate units per year, from ordinary least squares on eight annual values. The WHO standard population is applied in five-year bands, and the age-standardised rows for ages 7 or more use the 5–9 band at three fifths of its weight.

Specification | Rate 2014 | Rate 2021 | Change (%) | Slope | 95% CI | p

Ages ≥7, age-standardised | | | | | |

Brazil 2014 standard, 2018 projection (primary) | 1.123 | 1.078 | −4.0 | 0.0002 | (−0.0213 to 0.0217) | 0.983

Brazil 2014 standard, 2024 projection | 1.128 | 1.112 | −1.5 | 0.0040 | (−0.0176 to 0.0255) | 0.668

WHO standard, 2018 projection | 1.095 | 1.049 | −4.2 | −0.0001 | (−0.0207 to 0.0205) | 0.990

WHO standard, 2024 projection | 1.102 | 1.080 | −2.0 | 0.0031 | (−0.0176 to 0.0237) | 0.727

Ages 5–34 | | | | | |

Crude group rate, 2018 projection (primary) | 0.182 | 0.219 | +20.3 | 0.0080 | (0.0023 to 0.0138) | 0.014

Crude group rate, 2024 projection | 0.183 | 0.222 | +21.7 | 0.0085 | (0.0026 to 0.0144) | 0.012

WHO standard, 2018 projection | 0.178 | 0.208 | +16.9 | 0.0069 | (0.0015 to 0.0122) | 0.020

WHO standard, 2024 projection | 0.179 | 0.213 | +19.1 | 0.0075 | (0.0021 to 0.0130) | 0.015

Supplementary Table S6. Kitagawa decomposition of the change in the crude rate, completed ages 7 years and older (post hoc analysis). Rates and components per 100,000; population from the IBGE 2018 projection. The two components sum to the change in the crude rate. aAge-composition component as a percentage of the change in the crude rate; it exceeds 100% when the age-specific-rate component is negative, that is, when changes in age-specific rates partly offset the effect of the changing age composition.

Years compared | Crude rate, start / end | Change | Age composition | Age-specific rates | Composition share (%)a | Ages ≥60 in composition (%)

2014–2021 | 1.123 / 1.258 | +0.135 | +0.191 | −0.057 | 142 | 95

2014–2019 | 1.123 / 1.276 | +0.154 | +0.138 | +0.016 | 90 | 95

2014–2020 | 1.123 / 1.401 | +0.278 | +0.171 | +0.107 | 62 | 94

2015–2021 | 1.171 / 1.258 | +0.087 | +0.167 | −0.081 | 193 | 95

2016–2021 | 1.184 / 1.258 | +0.073 | +0.140 | −0.067 | 191 | 95

2014–2018 | 1.123 / 1.194 | +0.071 | +0.107 | −0.036 | 151 | 94

Supplementary Figure S1. Mortality at ages 5–34 years

Asthma mortality at ages 5–34 years, Brazil, 2014–2021. Upper panel: rate per 100,000 at ages 5–34 (solid line, filled circles) and at ages 7–34, the control group matching the age filter of the original analysis (dashed line, open squares). Lower panel: annual asthma deaths at ages 5–34. Population from the IBGE 2018 projection.

All analyses use corrected selection unless labelled historical. Rates and slope units are deaths per 100,000; APC is percent per year. Full-precision CSV results and model warnings are delivered with the code.

Table S1. Corrected selection by year

Year | All SIM | J45/J46 | COVID | Duplicates | Invalid dates | Negative age | Valid | ≥7 | 5–34

2014 | 1227039 | 2129 | 0 | 0 | 2 | 0 | 2127 | 2035 | 182

2015 | 1264175 | 2232 | 0 | 0 | 1 | 0 | 2231 | 2142 | 189

2016 | 1309774 | 2249 | 0 | 0 | 2 | 0 | 2247 | 2187 | 184

2017 | 1312663 | 2477 | 0 | 0 | 1 | 0 | 2476 | 2418 | 172

2018 | 1316719 | 2307 | 0 | 0 | 4 | 0 | 2303 | 2244 | 200

2019 | 1349801 | 2488 | 0 | 0 | 2 | 0 | 2486 | 2419 | 218

2020 | 1556824 | 2735 | 27 | 0 | 3 | 0 | 2705 | 2677 | 236

2021 | 1832649 | 2488 | 30 | 0 | 0 | 0 | 2458 | 2422 | 209

COVID = union across five cause lines. Invalid-date and negative-age counts follow COVID and deduplication. Both ambiguous 2018 records are retained. The historical selection, provided separately, removed 56 COVID records, one record under the reduced key and 15 records with invalid dates, leaving 18,583 deaths at continuous age >6.

Table S2. Annual Aligned crude ≥7

Year | Deaths | Population | Rate

2014 | 2035 | 181,252,457 | 1.122743

2015 | 2142 | 182,968,415 | 1.170694

2016 | 2187 | 184,647,932 | 1.184416

2017 | 2418 | 186,273,291 | 1.298093

2018 | 2244 | 187,895,171 | 1.194283

2019 | 2419 | 189,505,848 | 1.276478

2020 | 2677 | 191,083,309 | 1.400960

2021 | 2422 | 192,673,672 | 1.257048

Table S2. Annual Brazil 2014 ASR ≥7

Year | Deaths | Population | Rate

2014 | 2035 | 181,252,457 | 1.122743

2015 | 2142 | 182,968,415 | 1.144431

2016 | 2187 | 184,647,932 | 1.132194

2017 | 2418 | 186,273,291 | 1.208506

2018 | 2244 | 187,895,171 | 1.089633

2019 | 2419 | 189,505,848 | 1.140337

2020 | 2677 | 191,083,309 | 1.229760

2021 | 2422 | 192,673,672 | 1.077177

Table S2. Annual Crude 5–34

Year | Deaths | Population | Rate

2014 | 182 | 100,130,699 | 0.181762

2015 | 189 | 99,610,503 | 0.189739

2016 | 184 | 99,048,598 | 0.185767

2017 | 172 | 98,452,312 | 0.174704

2018 | 200 | 97,819,731 | 0.204458

2019 | 218 | 97,214,347 | 0.224247

2020 | 236 | 96,671,200 | 0.244126

2021 | 209 | 96,070,603 | 0.217548

Table S2. Annual Brazil 2014 ASR 5–34

Year | Deaths | Population | Rate

2014 | 182 | 100,130,699 | 0.181762

2015 | 189 | 99,610,503 | 0.189060

2016 | 184 | 99,048,598 | 0.184945

2017 | 172 | 98,452,312 | 0.174255

2018 | 200 | 97,819,731 | 0.203444

2019 | 218 | 97,214,347 | 0.222528

2020 | 236 | 96,671,200 | 0.241039

2021 | 209 | 96,070,603 | 0.214535

Table S2. Annual Crude 7–34

Year | Deaths | Population | Rate

2014 | 171 | 94,235,685 | 0.181460

2015 | 174 | 93,780,394 | 0.185540

2016 | 179 | 93,255,055 | 0.191947

2017 | 164 | 92,641,638 | 0.177026

2018 | 190 | 92,007,559 | 0.206505

2019 | 208 | 91,371,402 | 0.227642

2020 | 232 | 90,729,128 | 0.255706

2021 | 201 | 90,129,917 | 0.223011

Table S2. Annual Crude 35–59

Year | Deaths | Population | Rate

2014 | 452 | 62,893,940 | 0.718670

2015 | 485 | 64,149,669 | 0.756044

2016 | 515 | 65,398,428 | 0.787481

2017 | 567 | 66,640,781 | 0.850830

2018 | 504 | 67,862,310 | 0.742680

2019 | 524 | 69,039,371 | 0.758987

2020 | 709 | 70,157,104 | 1.010589

2021 | 591 | 71,213,520 | 0.829899

Table S2. Annual Crude ≥60

Year | Deaths | Population | Rate

2014 | 1412 | 24,122,832 | 5.853376

2015 | 1483 | 25,038,352 | 5.922914

2016 | 1493 | 25,994,449 | 5.743534

2017 | 1687 | 26,990,872 | 6.250261

2018 | 1550 | 28,025,302 | 5.530716

2019 | 1687 | 29,095,075 | 5.798232

2020 | 1736 | 30,197,077 | 5.748901

2021 | 1630 | 31,330,235 | 5.202642

Table S3. Historical age-group reproduction

series | n_2014 | n_2021 | rate_2014 | rate_2021 | pct_change_2014_2021 | slope | p_value

legacy_lt18 | 43 | 37 | 0.07649 | 0.06945 | -9.19335 | -0.00047 | 0.85236

legacy_18_59 | 580 | 756 | 0.47786 | 0.58735 | 22.91210 | 0.02125 | 0.04388

legacy_gte60 | 1412 | 1630 | 5.85338 | 5.20264 | -11.11724 | -0.07124 | 0.13500

Historical <18 numerator is floor ages 7–17 and denominator ages 0–17. These rows reproduce the historical construction and are not corrected primary domains.

Table S4. Count models, IBGE 2018

Series | Family | APC | 95% CI | p | Dispersion | Theta | Status

Crude ≥7 | Poisson | 2.190 | 1.549, 2.835 | <0.001 | 6.04 | | ok

Crude ≥7 | Quasi-Poisson | 2.190 | 0.236, 4.183 | 0.033 | 6.04 | | ok

Crude ≥7 | Negative binomial | 2.221 | 0.886, 3.574 | 0.001 | 1.33 | 686.8 | ok

Crude 5–34 | Poisson | 3.998 | 1.787, 6.258 | <0.001 | 1.09 | | ok

Crude 5–34 | Quasi-Poisson | 3.998 | 1.130, 6.949 | 0.014 | 1.09 | | ok

Crude 5–34 | Negative binomial | 3.998 | 1.787, 6.258 | <0.001 | 1.09 | 3280138.3 | warning

Crude 7–34 | Poisson | 4.604 | 2.327, 6.931 | <0.001 | 1.15 | | ok

Crude 7–34 | Quasi-Poisson | 4.604 | 1.561, 7.738 | 0.010 | 1.15 | | ok

Crude 7–34 | Negative binomial | 4.604 | 2.327, 6.931 | <0.001 | 1.15 | 2427388.1 | warning

Crude 35–59 | Poisson | 2.779 | 1.448, 4.126 | <0.001 | 5.54 | | ok

Crude 35–59 | Quasi-Poisson | 2.779 | -1.082, 6.789 | 0.130 | 5.54 | | ok

Crude 35–59 | Negative binomial | 2.783 | 0.181, 5.453 | 0.036 | 1.34 | 187.9 | ok

Crude ≥60 | Poisson | -1.300 | -2.047, -0.546 | <0.001 | 3.55 | | ok

Crude ≥60 | Quasi-Poisson | -1.300 | -3.049, 0.482 | 0.124 | 3.55 | | ok

Crude ≥60 | Negative binomial | -1.271 | -2.480, -0.046 | 0.042 | 1.34 | 972.6 | ok

Age-adjusted ≥7 | Poisson | 0.001 | -0.626, 0.632 | 0.998 | 3.11 | | ok

Age-adjusted ≥7 | Quasi-Poisson | 0.001 | -1.125, 1.140 | 0.999 | 3.11 | | ok

Age-adjusted ≥7 | Negative binomial | 0.926 | -0.197, 2.061 | 0.106 | 1.30 | 147.8 | ok

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

Table S4. Count models, IBGE 2024

Series | Family | APC | 95% CI | p | Dispersion | Theta | Status

Crude ≥7 | Poisson | 2.312 | 1.670, 2.958 | <0.001 | 5.98 | | ok

Crude ≥7 | Quasi-Poisson | 2.312 | 0.366, 4.297 | 0.027 | 5.98 | | ok

Crude ≥7 | Negative binomial | 2.342 | 1.012, 3.690 | <0.001 | 1.33 | 695.6 | ok

Crude 5–34 | Poisson | 4.178 | 1.964, 6.441 | <0.001 | 1.09 | | ok

Crude 5–34 | Quasi-Poisson | 4.178 | 1.303, 7.135 | 0.012 | 1.09 | | ok

Crude 5–34 | Negative binomial | 4.178 | 1.964, 6.441 | <0.001 | 1.09 | 3258238.9 | warning

Crude 7–34 | Poisson | 4.791 | 2.511, 7.122 | <0.001 | 1.16 | | ok

Crude 7–34 | Quasi-Poisson | 4.791 | 1.742, 7.931 | 0.008 | 1.16 | | ok

Crude 7–34 | Negative binomial | 4.791 | 2.511, 7.122 | <0.001 | 1.16 | 2402052.5 | warning

Crude 35–59 | Poisson | 2.843 | 1.511, 4.192 | <0.001 | 5.52 | | ok

Crude 35–59 | Quasi-Poisson | 2.843 | -1.013, 6.849 | 0.123 | 5.52 | | ok

Crude 35–59 | Negative binomial | 2.844 | 0.246, 5.510 | 0.032 | 1.34 | 189.0 | ok

Crude ≥60 | Poisson | -1.210 | -1.959, -0.455 | 0.002 | 3.33 | | ok

Crude ≥60 | Quasi-Poisson | -1.210 | -2.907, 0.517 | 0.136 | 3.33 | | ok

Crude ≥60 | Negative binomial | -1.186 | -2.359, 0.001 | 0.050 | 1.34 | 1080.7 | ok

Age-adjusted ≥7 | Poisson | 0.290 | -0.340, 0.924 | 0.368 | 3.02 | | ok

Age-adjusted ≥7 | Quasi-Poisson | 0.290 | -0.823, 1.415 | 0.606 | 3.02 | | ok

Age-adjusted ≥7 | Negative binomial | 1.137 | 0.031, 2.256 | 0.044 | 1.31 | 156.0 | ok

Rows marked as warnings reached the theta iteration limit; convergence of mean-model IRLS does not prove convergence of theta. Retain these estimates with caution. The recorded warning was “iteration limit reached”. All errors, warnings, iterations, coefficients and fitted objects are retained in CSV/RDS. Quasi-Poisson uses t intervals with residual degrees of freedom; Poisson and negative binomial use normal Wald intervals.

Table S5. Standards and domains, IBGE 2018

Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p

Brazil ASR 35–59 | 0.71867 | 0.82485 | +0.10618 | +14.77 | 0.02134 (-0.00957, 0.05225) | 0.142

Brazil ASR 5–34 | 0.18176 | 0.21453 | +0.03277 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018

Brazil ASR 7–34 | 0.18146 | 0.21955 | +0.03809 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013

Brazil ASR ≥60 | 5.85338 | 5.08539 | -0.76799 | -13.12 | -0.08864 (-0.18714, 0.00986) | 0.070

Brazil ASR ≥7 | 1.12274 | 1.07718 | -0.04557 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986

WHO ASR 35–59 | 0.71921 | 0.82668 | +0.10747 | +14.94 | 0.02162 (-0.00936, 0.05260) | 0.139

WHO ASR 5–34 | 0.17788 | 0.20702 | +0.02914 | +16.38 | 0.00678 (0.00140, 0.01216) | 0.022

WHO ASR 7–34 | 0.17730 | 0.21246 | +0.03516 | +19.83 | 0.00819 (0.00235, 0.01403) | 0.014

WHO ASR ≥60 | 5.55943 | 4.81550 | -0.74393 | -13.38 | -0.08637 (-0.17725, 0.00451) | 0.059

WHO ASR ≥7 | 1.09451 | 1.04795 | -0.04656 | -4.25 | -0.00015 (-0.02077, 0.02047) | 0.986

Table S5. Standards and domains, IBGE 2024

Series | 2014 | 2021 | Δ rate | Δ % | Slope (95% CI) | p

Brazil ASR 35–59 | 0.72112 | 0.83140 | +0.11028 | +15.29 | 0.02192 (-0.00907, 0.05291) | 0.134

Brazil ASR 5–34 | 0.18281 | 0.21972 | +0.03691 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014

Brazil ASR 7–34 | 0.18258 | 0.22505 | +0.04247 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011

Brazil ASR ≥60 | 5.88365 | 5.30051 | -0.58314 | -9.91 | -0.06471 (-0.16221, 0.03278) | 0.155

Brazil ASR ≥7 | 1.12844 | 1.11123 | -0.01722 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671

WHO ASR 35–59 | 0.72095 | 0.83282 | +0.11187 | +15.52 | 0.02223 (-0.00879, 0.05326) | 0.130

WHO ASR 5–34 | 0.17884 | 0.21201 | +0.03317 | +18.55 | 0.00744 (0.00193, 0.01294) | 0.016

WHO ASR 7–34 | 0.17835 | 0.21778 | +0.03943 | +22.11 | 0.00889 (0.00290, 0.01487) | 0.011

WHO ASR ≥60 | 5.60811 | 5.01048 | -0.59763 | -10.66 | -0.06712 (-0.15761, 0.02337) | 0.119

WHO ASR ≥7 | 1.10228 | 1.07940 | -0.02288 | -2.08 | 0.00305 (-0.01761, 0.02370) | 0.731

Table S6. Kitagawa, IBGE 2018

Endpoints | Crude start | Crude end | Net Δ | Composition | Rate term | Composition %

2014–2021 | 1.122743 | 1.257048 | +0.134304 | +0.191434 | -0.057129 | 142.54

2014–2019 | 1.122743 | 1.276478 | +0.153734 | +0.138122 | +0.015612 | 89.84

2014–2020 | 1.122743 | 1.400960 | +0.278216 | +0.171157 | +0.107059 | 61.52

2015–2021 | 1.170694 | 1.257048 | +0.086354 | +0.167391 | -0.081037 | 193.84

2016–2021 | 1.184416 | 1.257048 | +0.072632 | +0.139935 | -0.067304 | 192.66

2014–2018 | 1.122743 | 1.194283 | +0.071540 | +0.107366 | -0.035827 | 150.08

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

Table S6. Kitagawa, IBGE 2024

Endpoints | Crude start | Crude end | Net Δ | Composition | Rate term | Composition %

2014–2021 | 1.128443 | 1.274118 | +0.145675 | +0.170443 | -0.024768 | 117.00

2014–2019 | 1.128443 | 1.290082 | +0.161639 | +0.130868 | +0.030771 | 80.96

2014–2020 | 1.128443 | 1.417122 | +0.288679 | +0.158796 | +0.129883 | 55.01

2015–2021 | 1.177572 | 1.274118 | +0.096546 | +0.148180 | -0.051634 | 153.48

2016–2021 | 1.192574 | 1.274118 | +0.081544 | +0.121796 | -0.040252 | 149.36

2014–2018 | 1.128443 | 1.205894 | +0.077451 | +0.101345 | -0.023893 | 130.85

Each term is independently rounded. The unrounded identity is tested to 1e−12; rounded terms may differ in their last displayed digit. A share over 100% reflects a negative rate term, not a probability.

Table S7. Pandemic-period influence, age_5_34, IBGE 2018

Scenario | Series | Years | Δ % | Slope (95% CI) | p

Principal | Brazil ASR 5–34 | 8 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018

Principal | Crude 5–34 | 8 | +19.69 | 0.00795 (0.00212, 0.01377) | 0.016

Principal | WHO ASR 5–34 | 8 | +16.38 | 0.00678 (0.00140, 0.01216) | 0.022

Without 2020 | Brazil ASR 5–34 | 7 | +18.03 | 0.00581 (0.00040, 0.01122) | 0.040

Without 2020 | Crude 5–34 | 7 | +19.69 | 0.00619 (0.00070, 0.01167) | 0.034

Without 2020 | WHO ASR 5–34 | 7 | +16.38 | 0.00536 (-0.00017, 0.01089) | 0.055

2014–2019 | Brazil ASR 5–34 | 6 | +22.43 | 0.00675 (-0.00228, 0.01578) | 0.107

2014–2019 | Crude 5–34 | 6 | +23.37 | 0.00701 (-0.00219, 0.01622) | 0.102

2014–2019 | WHO ASR 5–34 | 6 | +22.80 | 0.00660 (-0.00250, 0.01570) | 0.114

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_5_34, IBGE 2018

Scenario | Family | APC | 95% CI | p | df | Status

Principal | Poisson | 3.998 | 1.787, 6.258 | <0.001 | 6 | ok

Principal | Quasi-Poisson | 3.998 | 1.130, 6.949 | 0.014 | 6 | ok

Principal | Negative binomial | 3.998 | 1.787, 6.258 | <0.001 | 6 | warning

Without 2020 | Poisson | 3.159 | 0.743, 5.633 | 0.010 | 5 | ok

Without 2020 | Quasi-Poisson | 3.159 | 0.382, 6.012 | 0.033 | 5 | ok

Without 2020 | Negative binomial | 3.159 | 0.743, 5.633 | 0.010 | 5 | warning

2014–2019 | Poisson | 3.673 | 0.212, 7.253 | 0.037 | 4 | ok

2014–2019 | Quasi-Poisson | 3.673 | -0.992, 8.558 | 0.095 | 4 | ok

2014–2019 | Negative binomial | 3.673 | 0.212, 7.253 | 0.037 | 4 | warning

Table S7. Pandemic-period influence, age_7_34, IBGE 2018

Scenario | Series | Years | Δ % | Slope (95% CI) | p

Principal | Brazil ASR 7–34 | 8 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013

Principal | Crude 7–34 | 8 | +22.90 | 0.00926 (0.00294, 0.01559) | 0.012

Principal | WHO ASR 7–34 | 8 | +19.83 | 0.00819 (0.00235, 0.01403) | 0.014

Without 2020 | Brazil ASR 7–34 | 7 | +20.99 | 0.00666 (0.00152, 0.01180) | 0.021

Without 2020 | Crude 7–34 | 7 | +22.90 | 0.00710 (0.00188, 0.01232) | 0.017

Without 2020 | WHO ASR 7–34 | 7 | +19.83 | 0.00633 (0.00108, 0.01158) | 0.027

2014–2019 | Brazil ASR 7–34 | 6 | +24.38 | 0.00767 (-0.00086, 0.01620) | 0.067

2014–2019 | Crude 7–34 | 6 | +25.45 | 0.00797 (-0.00076, 0.01670) | 0.064

2014–2019 | WHO ASR 7–34 | 6 | +25.20 | 0.00768 (-0.00087, 0.01622) | 0.067

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_7_34, IBGE 2018

Scenario | Family | APC | 95% CI | p | df | Status

Principal | Poisson | 4.604 | 2.327, 6.931 | <0.001 | 6 | ok

Principal | Quasi-Poisson | 4.604 | 1.561, 7.738 | 0.010 | 6 | ok

Principal | Negative binomial | 4.604 | 2.327, 6.931 | <0.001 | 6 | warning

Without 2020 | Poisson | 3.590 | 1.104, 6.138 | 0.004 | 5 | ok

Without 2020 | Quasi-Poisson | 3.590 | 0.975, 6.274 | 0.016 | 5 | ok

Without 2020 | Negative binomial | 3.590 | 1.104, 6.138 | 0.004 | 5 | warning

2014–2019 | Poisson | 4.153 | 0.584, 7.849 | 0.022 | 4 | ok

2014–2019 | Quasi-Poisson | 4.153 | -0.220, 8.718 | 0.058 | 4 | ok

2014–2019 | Negative binomial | 4.153 | 0.583, 7.849 | 0.022 | 4 | warning

Table S7. Pandemic-period influence, age_5_34, IBGE 2024

Scenario | Series | Years | Δ % | Slope (95% CI) | p

Principal | Brazil ASR 5–34 | 8 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014

Principal | Crude 5–34 | 8 | +21.07 | 0.00841 (0.00249, 0.01433) | 0.013

Principal | WHO ASR 5–34 | 8 | +18.55 | 0.00744 (0.00193, 0.01294) | 0.016

Without 2020 | Brazil ASR 5–34 | 7 | +20.19 | 0.00643 (0.00087, 0.01199) | 0.031

Without 2020 | Crude 5–34 | 7 | +21.07 | 0.00662 (0.00105, 0.01220) | 0.028

Without 2020 | WHO ASR 5–34 | 7 | +18.55 | 0.00596 (0.00036, 0.01155) | 0.041

2014–2019 | Brazil ASR 5–34 | 6 | +24.27 | 0.00740 (-0.00188, 0.01669) | 0.091

2014–2019 | Crude 5–34 | 6 | +24.69 | 0.00754 (-0.00180, 0.01687) | 0.088

2014–2019 | WHO ASR 5–34 | 6 | +24.37 | 0.00720 (-0.00201, 0.01642) | 0.096

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_5_34, IBGE 2024

Scenario | Family | APC | 95% CI | p | df | Status

Principal | Poisson | 4.178 | 1.964, 6.441 | <0.001 | 6 | ok

Principal | Quasi-Poisson | 4.178 | 1.303, 7.135 | 0.012 | 6 | ok

Principal | Negative binomial | 4.178 | 1.964, 6.441 | <0.001 | 6 | warning

Without 2020 | Poisson | 3.339 | 0.919, 5.816 | 0.007 | 5 | ok

Without 2020 | Quasi-Poisson | 3.339 | 0.551, 6.203 | 0.027 | 5 | ok

Without 2020 | Negative binomial | 3.339 | 0.919, 5.816 | 0.007 | 5 | warning

2014–2019 | Poisson | 3.899 | 0.431, 7.487 | 0.027 | 4 | ok

2014–2019 | Quasi-Poisson | 3.899 | -0.768, 8.786 | 0.082 | 4 | ok

2014–2019 | Negative binomial | 3.899 | 0.431, 7.487 | 0.027 | 4 | warning

Table S7. Pandemic-period influence, age_7_34, IBGE 2024

Scenario | Series | Years | Δ % | Slope (95% CI) | p

Principal | Brazil ASR 7–34 | 8 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011

Principal | Crude 7–34 | 8 | +24.37 | 0.00977 (0.00334, 0.01620) | 0.010

Principal | WHO ASR 7–34 | 8 | +22.11 | 0.00889 (0.00290, 0.01487) | 0.011

Without 2020 | Brazil ASR 7–34 | 7 | +23.26 | 0.00732 (0.00202, 0.01262) | 0.016

Without 2020 | Crude 7–34 | 7 | +24.37 | 0.00757 (0.00225, 0.01288) | 0.015

Without 2020 | WHO ASR 7–34 | 7 | +22.11 | 0.00696 (0.00164, 0.01227) | 0.020

2014–2019 | Brazil ASR 7–34 | 6 | +26.32 | 0.00837 (-0.00043, 0.01716) | 0.057

2014–2019 | Crude 7–34 | 6 | +26.85 | 0.00853 (-0.00033, 0.01739) | 0.056

2014–2019 | WHO ASR 7–34 | 6 | +26.85 | 0.00831 (-0.00035, 0.01697) | 0.056

Without 2020 uses 2014–2019 and 2021 as calendar years (seven observations, 5 residual df); prepandemic uses 2014–2019 (six observations, 4 residual df). Endpoint contrasts are distinct from annual slopes. Count-model sensitivities for the same domains are supplied below.

Table S7 continued. Count models, age_7_34, IBGE 2024

Scenario | Family | APC | 95% CI | p | df | Status

Principal | Poisson | 4.791 | 2.511, 7.122 | <0.001 | 6 | ok

Principal | Quasi-Poisson | 4.791 | 1.742, 7.931 | 0.008 | 6 | ok

Principal | Negative binomial | 4.791 | 2.511, 7.122 | <0.001 | 6 | warning

Without 2020 | Poisson | 3.778 | 1.288, 6.329 | 0.003 | 5 | ok

Without 2020 | Quasi-Poisson | 3.778 | 1.150, 6.474 | 0.014 | 5 | ok

Without 2020 | Negative binomial | 3.778 | 1.288, 6.329 | 0.003 | 5 | warning

2014–2019 | Poisson | 4.391 | 0.814, 8.096 | 0.016 | 4 | ok

2014–2019 | Quasi-Poisson | 4.391 | 0.018, 8.956 | 0.049 | 4 | ok

2014–2019 | Negative binomial | 4.391 | 0.814, 8.096 | 0.016 | 4 | warning

Table S8. One-condition sensitivities, ge7, IBGE 2018

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 18544 | 1.12274 | 1.07718 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986

Days/365.25 | 18544 | 1.12274 | 1.07719 | -4.06 | 0.00016 (-0.02139, 0.02170) | 0.986

Original COVID | 18545 | 1.12274 | 1.07772 | -4.01 | 0.00020 (-0.02130, 0.02171) | 0.982

Retain COVID | 18601 | 1.12274 | 1.09149 | -2.78 | 0.00210 (-0.01965, 0.02385) | 0.821

Historical key | 18543 | 1.12274 | 1.07718 | -4.06 | 0.00015 (-0.02142, 0.02172) | 0.987

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_5_34, IBGE 2018

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018

Days/365.25 | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018

Original COVID | 1591 | 0.18176 | 0.21553 | +18.58 | 0.00760 (0.00195, 0.01324) | 0.017

Retain COVID | 1596 | 0.18176 | 0.21960 | +20.82 | 0.00800 (0.00246, 0.01353) | 0.012

Historical key | 1590 | 0.18176 | 0.21453 | +18.03 | 0.00751 (0.00181, 0.01322) | 0.018

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_7_34, IBGE 2018

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013

Days/365.25 | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013

Original COVID | 1520 | 0.18146 | 0.22060 | +21.57 | 0.00884 (0.00275, 0.01493) | 0.012

Retain COVID | 1525 | 0.18146 | 0.22493 | +23.96 | 0.00926 (0.00331, 0.01522) | 0.009

Historical key | 1519 | 0.18146 | 0.21955 | +20.99 | 0.00875 (0.00260, 0.01490) | 0.013

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, ge7, IBGE 2024

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 18544 | 1.12844 | 1.11123 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671

Days/365.25 | 18544 | 1.12844 | 1.11123 | -1.53 | 0.00393 (-0.01762, 0.02547) | 0.671

Original COVID | 18545 | 1.12844 | 1.11178 | -1.48 | 0.00397 (-0.01753, 0.02547) | 0.667

Retain COVID | 18601 | 1.12844 | 1.12583 | -0.23 | 0.00590 (-0.01590, 0.02769) | 0.532

Historical key | 18543 | 1.12844 | 1.11123 | -1.53 | 0.00392 (-0.01765, 0.02549) | 0.672

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_5_34, IBGE 2024

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014

Days/365.25 | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014

Original COVID | 1591 | 0.18281 | 0.22073 | +20.74 | 0.00827 (0.00246, 0.01409) | 0.013

Retain COVID | 1596 | 0.18281 | 0.22498 | +23.06 | 0.00869 (0.00299, 0.01439) | 0.010

Historical key | 1590 | 0.18281 | 0.21972 | +20.19 | 0.00819 (0.00232, 0.01406) | 0.014

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Table S8. One-condition sensitivities, age_7_34, IBGE 2024

Scenario | Deaths | 2014 | 2021 | Δ % | Slope (95% CI) | p

Principal | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011

Days/365.25 | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011

Original COVID | 1520 | 0.18258 | 0.22612 | +23.85 | 0.00956 (0.00329, 0.01583) | 0.010

Retain COVID | 1525 | 0.18258 | 0.23063 | +26.32 | 0.01000 (0.00387, 0.01614) | 0.007

Historical key | 1519 | 0.18258 | 0.22505 | +23.26 | 0.00947 (0.00313, 0.01581) | 0.011

Each scenario changes only the indicated condition from corrected selection. Calendar-boundary changes are recorded individually in record_selection_and_age_transitions.csv; COVID overlaps in covid_overlap_patterns.csv.

Supplementary Figure S1

Figure S1. Annual crude rates for ages 5–34 and 7–34, within-5–34 Brazil-2014-standardised rates, and 5–34 death counts; corrected selection and IBGE 2018.

Supplementary methods and provenance

The historical deduplication key combined parsed death date, elapsed-days age, underlying cause and the five cause lines. It collapsed 2018 source rows 971828 and 971831. Inspection of all public fields found differences in occupation and seven administrative fields (NUMEROLOTE, TPPOS, DTCADASTRO, DTRECEBIM, DTRECORIGA, DIFDATA and CONTADOR). The SIM dictionary identifies NUMERODO as the death-declaration number and describes registration and receipt dates as administrative information [6]. NUMERODO is absent from these CSVs. The available fields therefore do not establish whether the rows refer to the same death. We retained both in the primary analysis and assessed their collapse under the historical key as a sensitivity.

Completed age was the difference between death and birth years, reduced by one if death preceded that year's birthday. For births on 29 February, the anniversary in a non-leap year was 1 March. We added a leading zero to seven-digit day-month-year strings and required an eight-digit valid calendar date. Missing or uninterpretable dates and birth after death were excluded and logged separately. We applied no upper-age exclusion and grouped ages ≥90 into the IBGE 90+ interval.

We reconciled the archived 12-page manuscript of 10 October 2026 with a Word copy of concordant content and repository commit 3884c868a07164710a2e358180b3034f9cdcf642. Official SIM archives recovered on 11 October 2026 reproduced the versioned selection counts and year-age aggregates exactly. Historical execution hashes were available for SIM 2020–2021 only, so byte identity with the historical 2014–2019 files could not be established. The accompanying provenance manifest records the reference PDF hash and all verified input hashes.

National single-year populations for both sexes came from the Brazilian Institute of Geography and Statistics (IBGE) 2018 revision [7]; the 2024 revision following the 2022 Census provided a sensitivity [8]. We used the total-population block on sheet BR in the 2018 workbook and SIGLA = BR with SEXO = Ambos in the 2024 workbook. Both provide ages 0–89 and an open 90+ interval. Corrected crude rates matched each numerator to the same population age range. To reproduce the historical construction, we divided continuous-age >6 deaths by the entire

population and separately by population ≥7. The latter comparator retains the small mismatch between its historical numerator and denominator.

We used R 4.6.1 [12] for reproduction and revision; the reference reported R 4.3.3. The code records package versions, configurations, input hashes, warnings and independent checks, including numerical differences under the current environment. Public inputs, verified extraction code and separate scenario outputs support reproduction. Code and revision outputs are available at https://github.com/EduIntellect/asthma-mortality-brazil-reanalysis/tree/revision/amaury-review .